# How Yes / No answers change during fitting

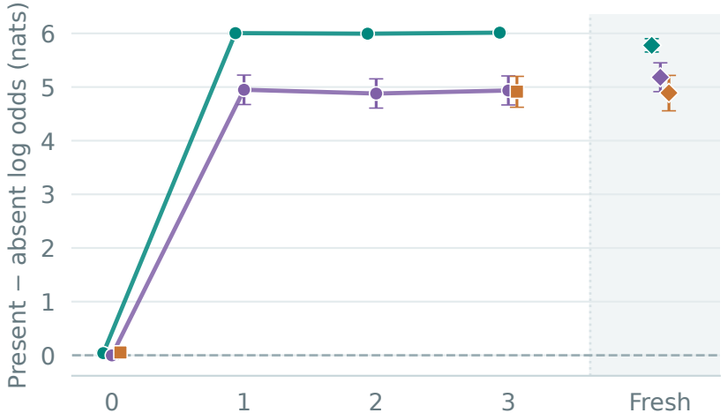
Qwen3-8B · only answer-token weights are fitted · internal model weights fixed

Three ridge corrections, not SFT updates. Fresh: both row edits installed, new compilation.

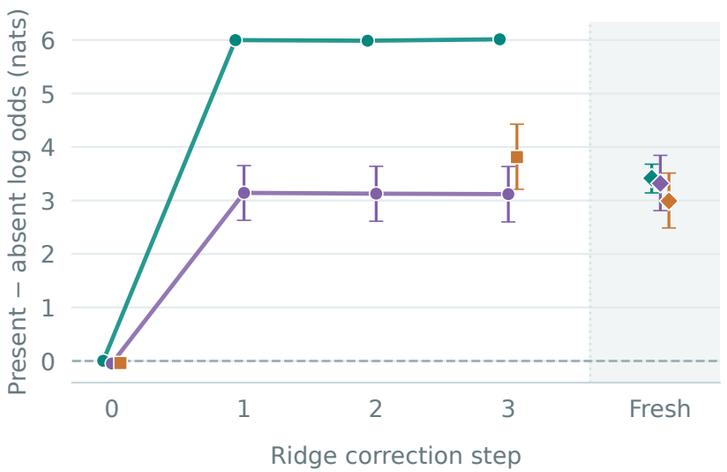
Training  Validation  Unseen test contexts

## Condition separation

### Gradient recording vs no-grad

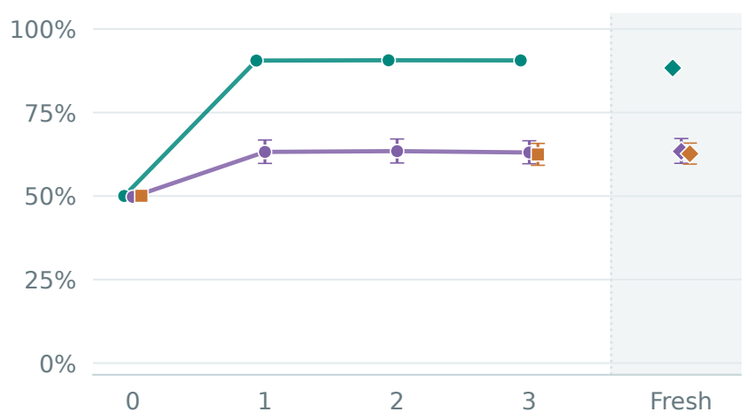



### Read-only probe vs no probe

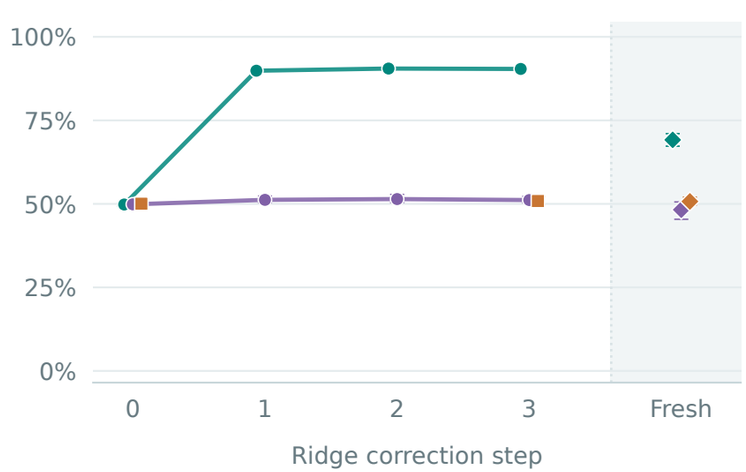



## Correct-token probability

### Gradient recording vs no-grad

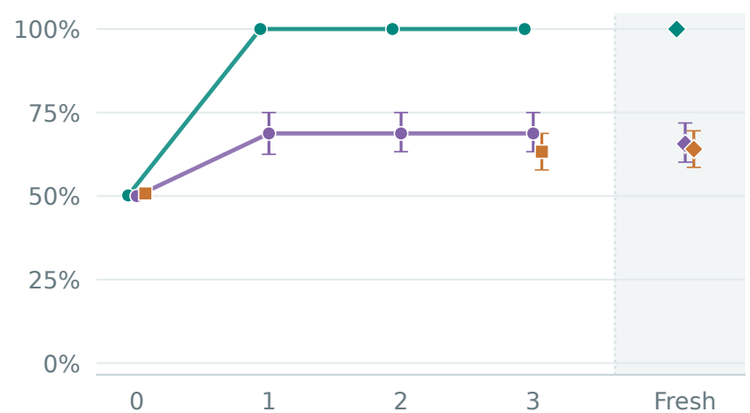



### Read-only probe vs no probe

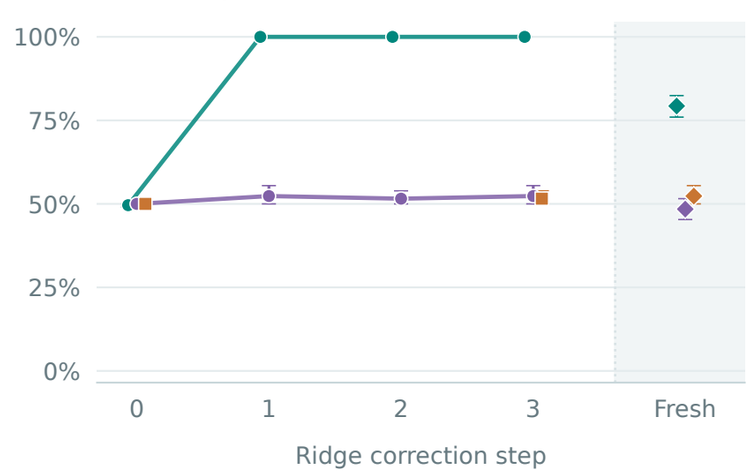



## Actual top-token accuracy

### Gradient recording vs no-grad




### Read-only probe vs no probe




Target tokens: No when absent, Yes when present. Probability and accuracy use the entire vocabulary.

Bars: pointwise 95% intervals from 5,000 whole-prompt bootstrap resamples; conditions and prompt order stay paired.

Intervals are not simultaneous and exclude training-seed / hardware variation. Test points are endpoints only; no test trend is inferred.

At 100% or 0% accuracy, empirical bootstrap intervals may collapse; this does not establish a population certainty.

# How animal answers change during fitting

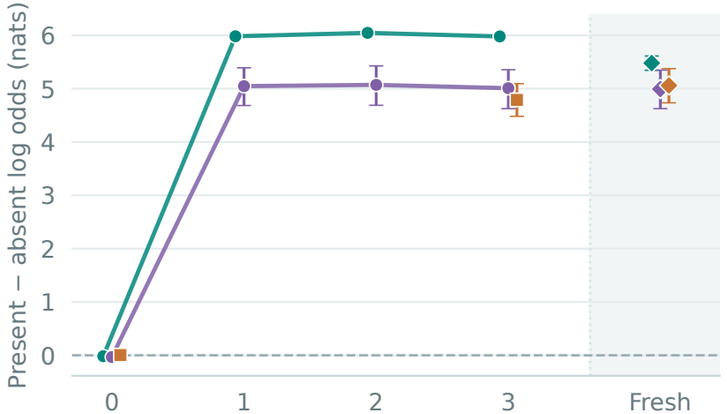
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Training  Validation  Unseen test contexts

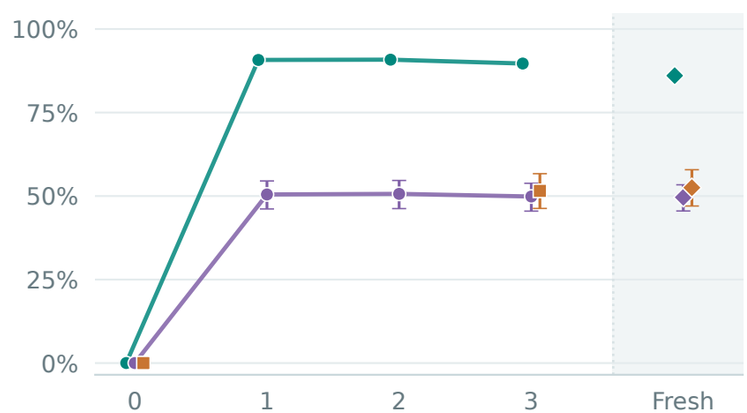
## Condition separation

### Gradient recording vs no-grad



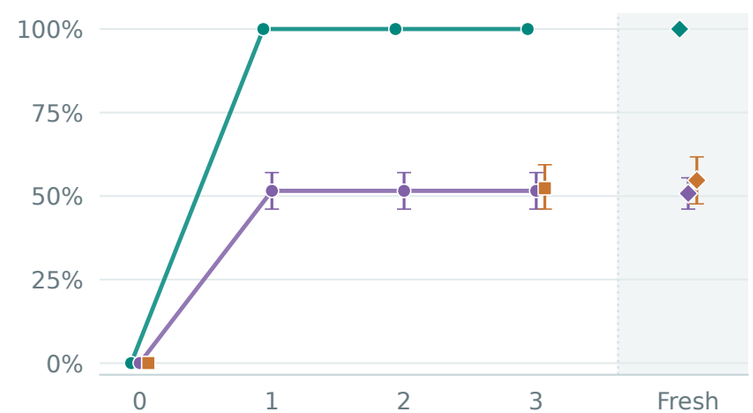

## Correct-token probability

### Gradient recording vs no-grad

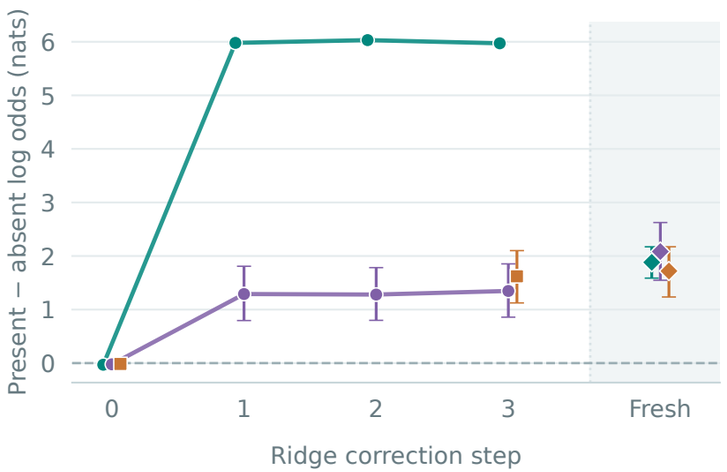



## Actual top-token accuracy

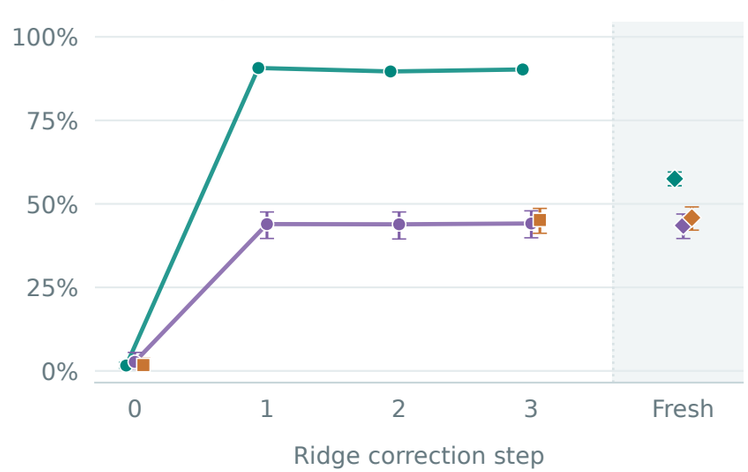
### Gradient recording vs no-grad



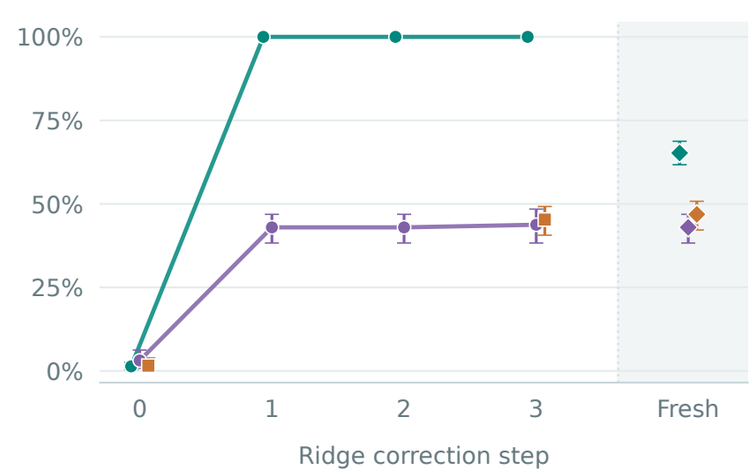

### Read-only probe vs no probe




### Read-only probe vs no probe




### Read-only probe vs no probe




Target tokens: owl when absent, cat when present. Probability and accuracy use the entire vocabulary.

Bars: pointwise 95% intervals from 5,000 whole-prompt bootstrap resamples; conditions and prompt order stay paired.

Intervals are not simultaneous and exclude training-seed / hardware variation. Test points are endpoints only; no test trend is inferred.

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